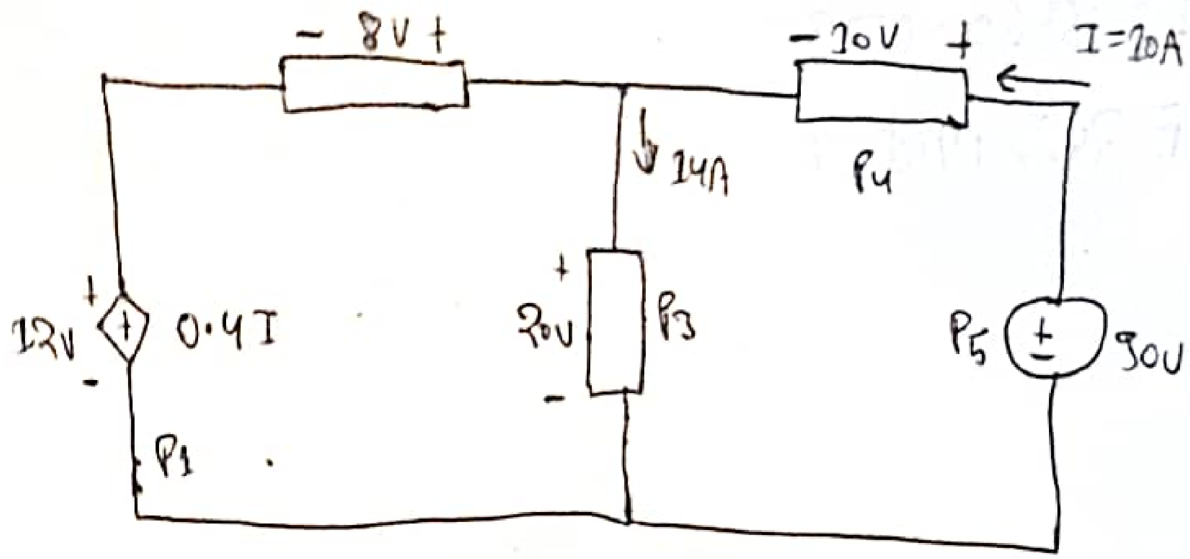
Oli Ahmed

2019-3-60-073

CSE-209, Mid-1

Ans to the ques no - 1



We know,

$$P = VI$$

Here,

$$P_1 = -(12 \times 0.4I)$$

$$= -(12 \times (0.4 \times 10))$$

$$= -48 \text{ watt}$$

(supply)

$P_2 = -(8 \times 4)$

$= -32$ watt

(supply)

$P_3 = 20 \times 14$

$= 280$ watt

(absorve)

$P_4 = 10 \times 10$

$= 100$ watt (absorve).

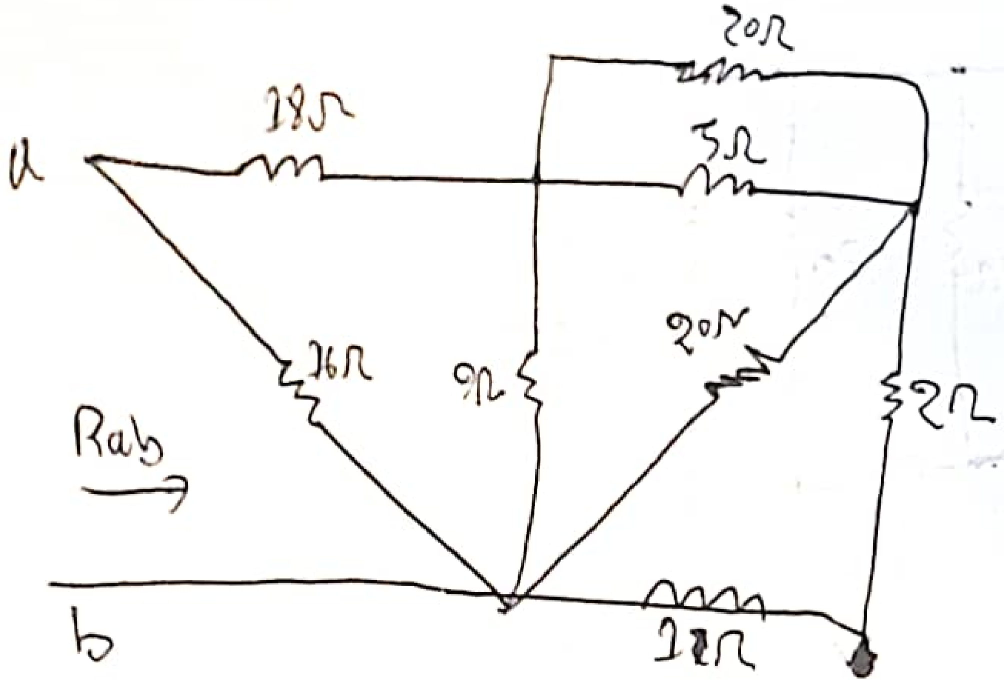
$P_5 = -(30 \times 10)$

$= -300$ watt

(supply).

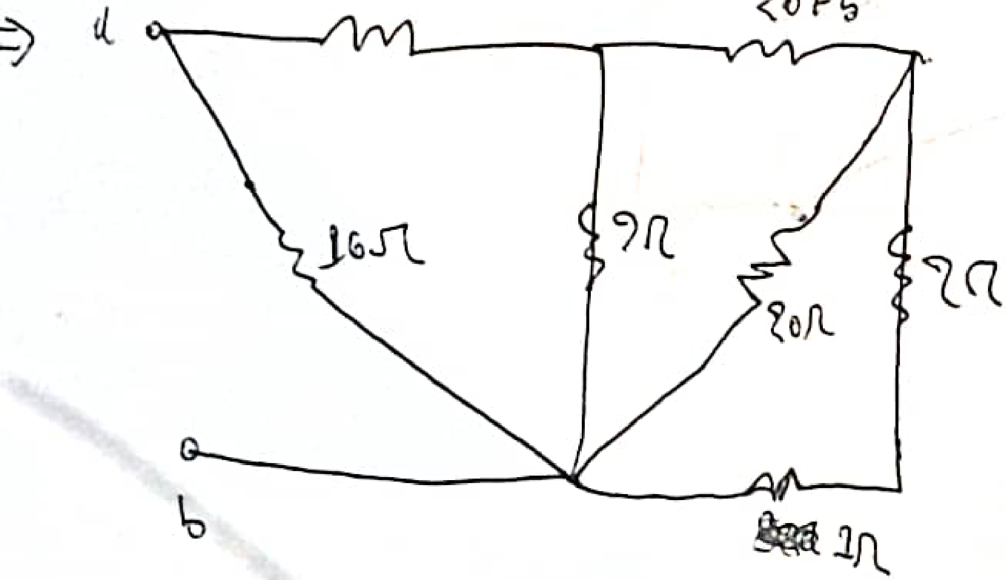
Ans

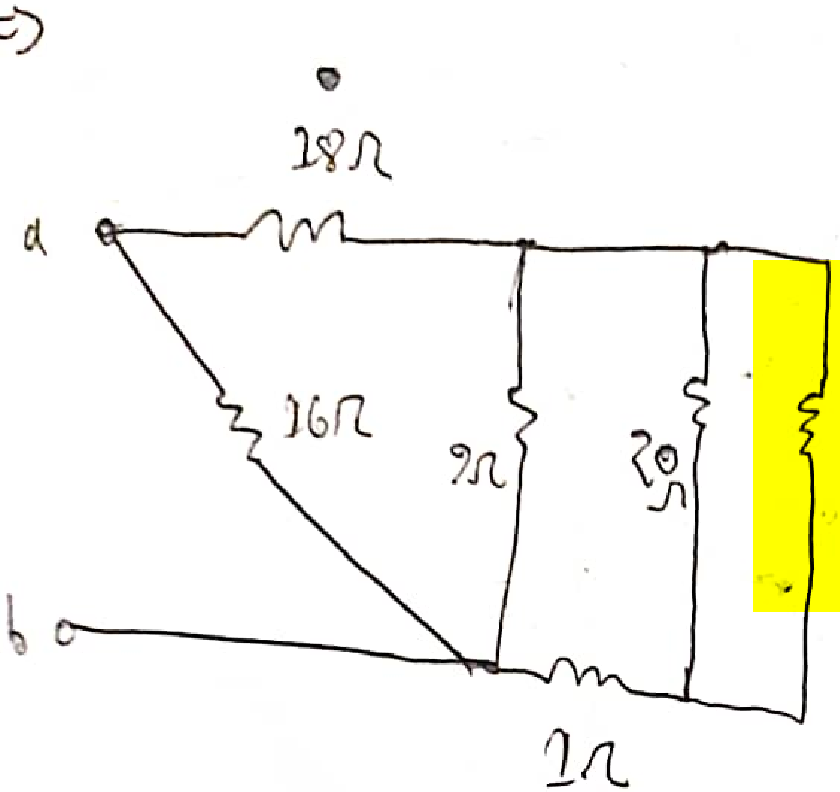
Ans to the ques no-2



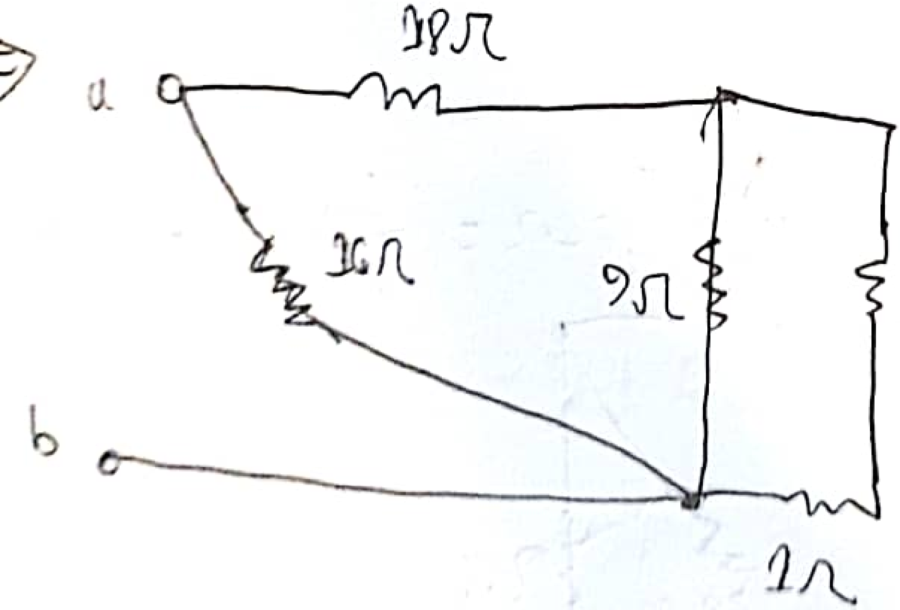
$$\frac{20\times5}{20+5}\,\Omega = \frac{100}{25}\,\Omega = 4\,\Omega$$

$\Rightarrow$

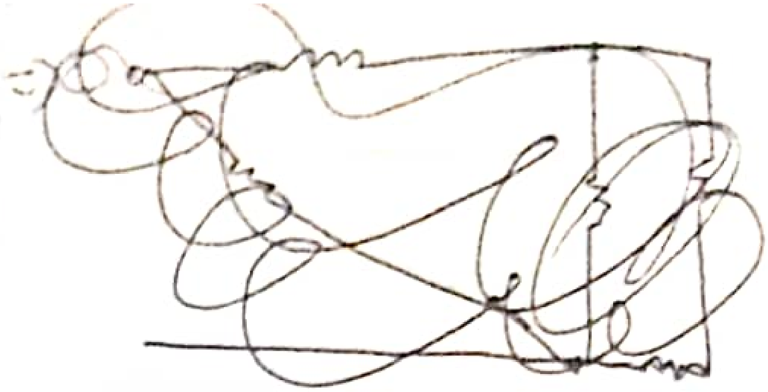




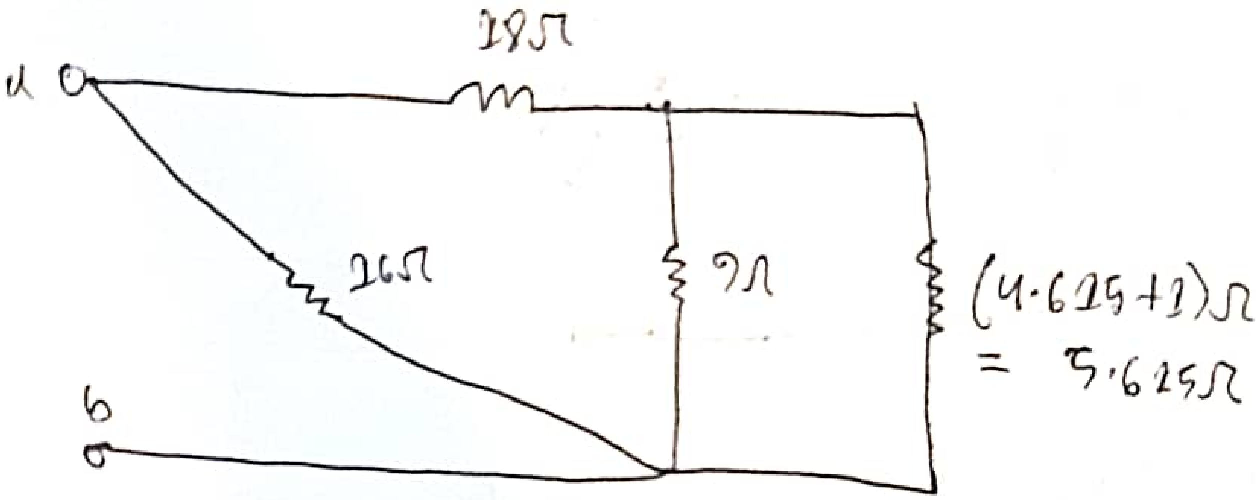
$(4+2) = 6\,\Omega$



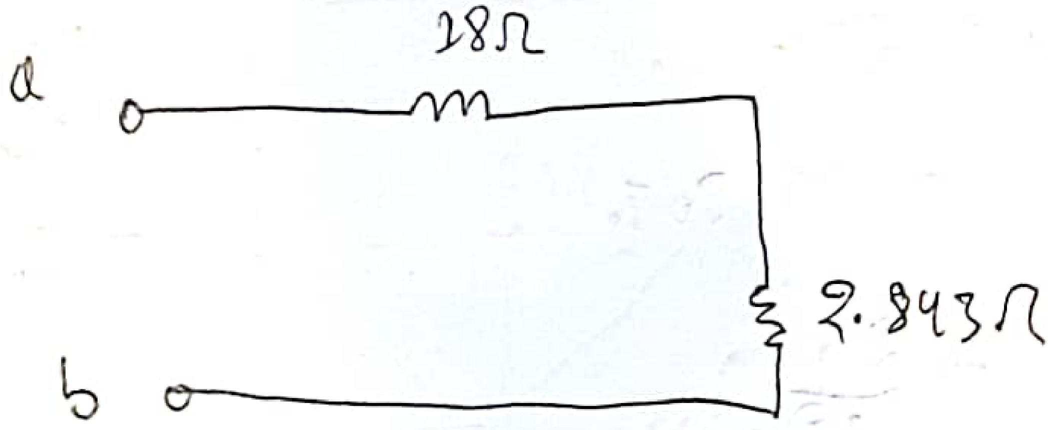
$$\frac{20\times 6}{20+6} = 4.615\,\Omega$$



$\Rightarrow$



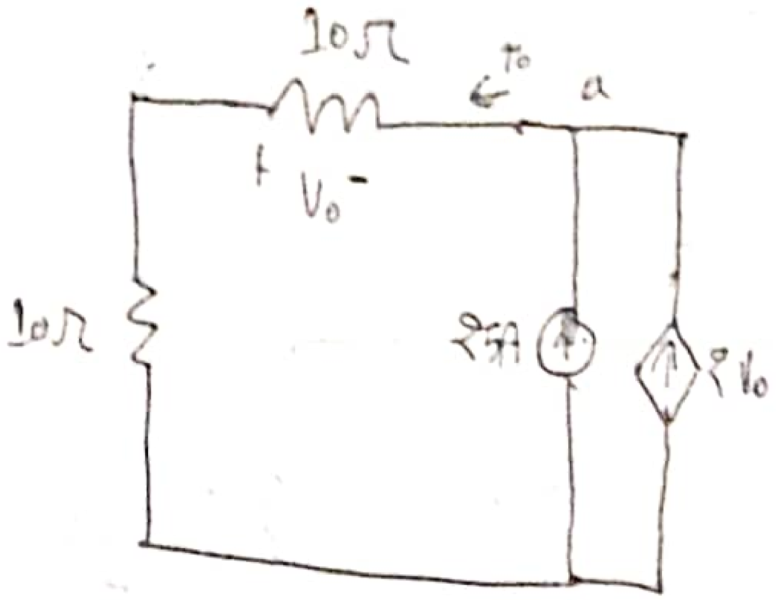
$\Rightarrow$



$\therefore R_b = (18+2.843)\Omega = 20.843\,\Omega$

Ans.

Ans to the ques no-3



Applying kcl at node a,

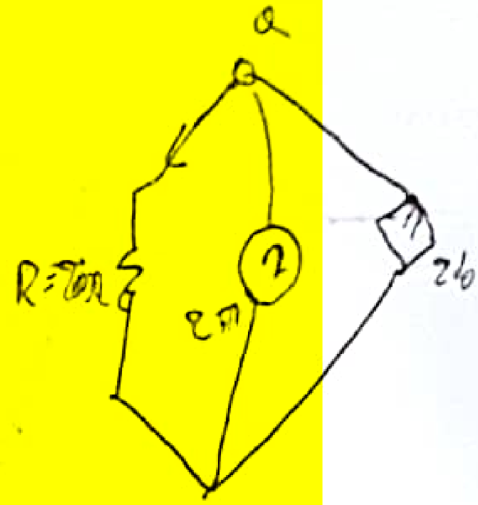
$$I_o = 25 + 2V_o$$
$$= 25 + 20 I_o$$
$$I_o = -\frac{25}{19}$$
$$= -1.31$$

Here,
$$V_o = I_o R$$
$$= I_o \times 20$$

$\therefore V_0 = I_0 R$

$= -1.31 \times 10$

$= -13.1 V$



Here,

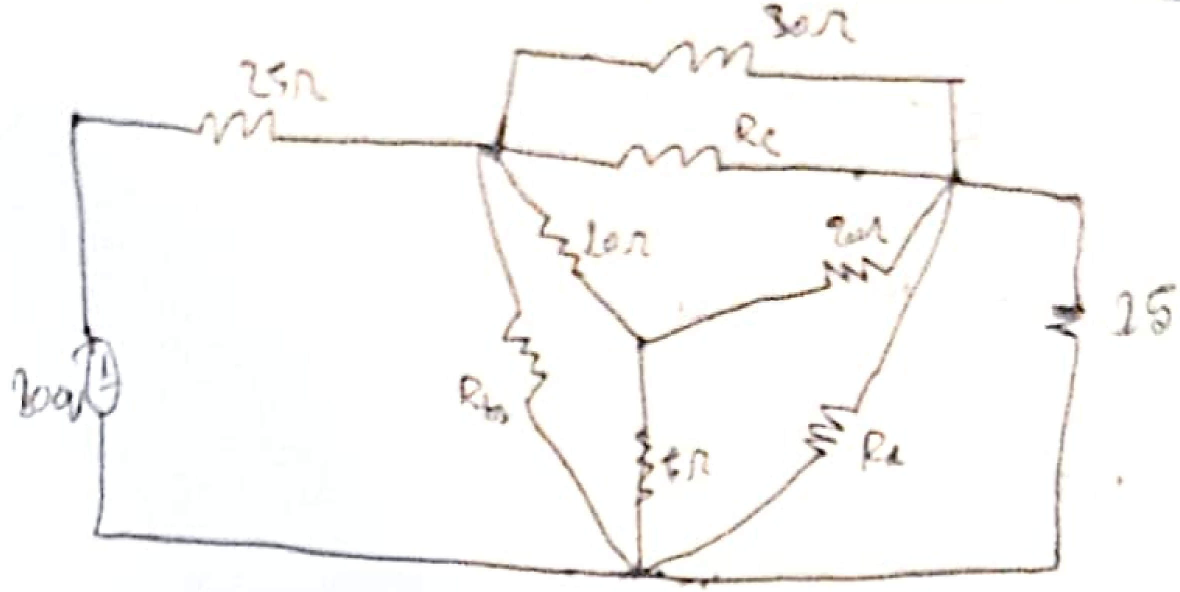
$V = I_0 \times R$

$= 1.3 \times 20$

$= -26.5$

Ans

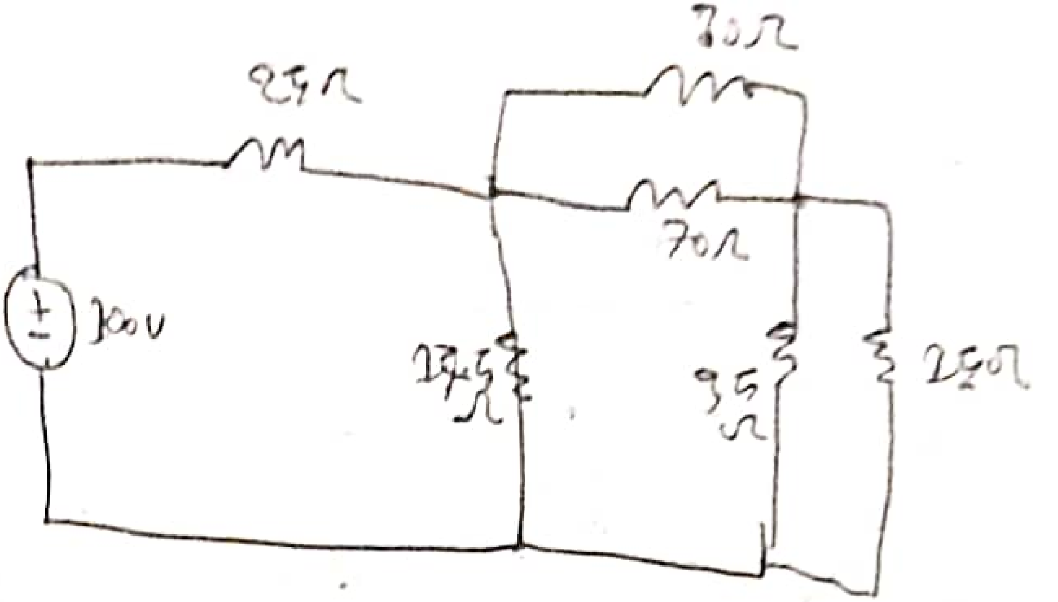
Ans. to the ques -4

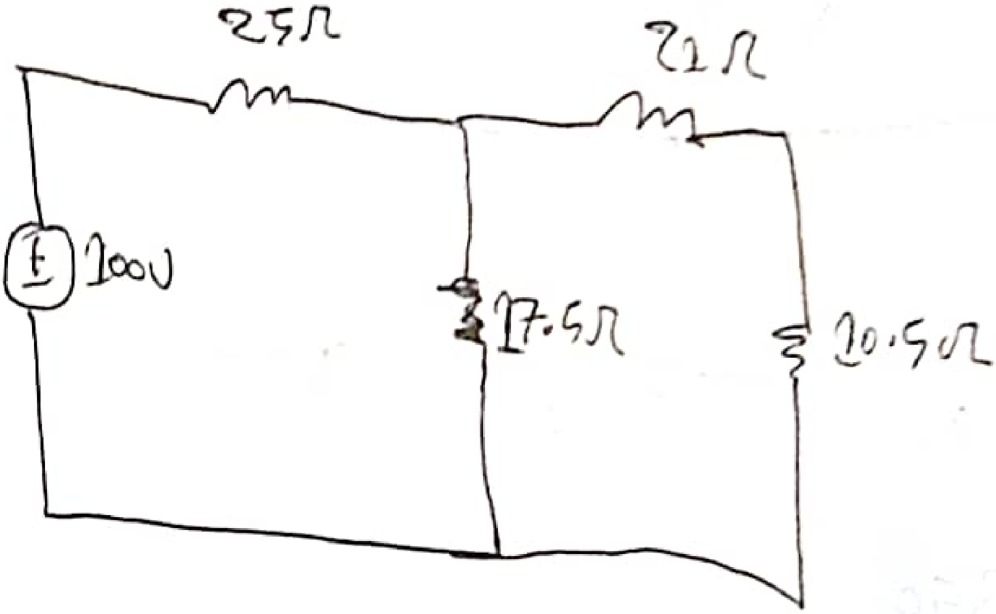


$$R_a = \frac{200+100+50}{10} = \frac{350}{10} = 35\Omega$$

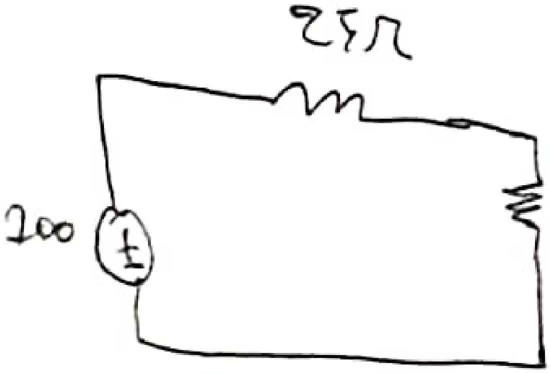
$$R_b = \frac{350}{20} = 17.5\Omega$$

$$R_c = \frac{350}{5} = 70\Omega$$





⇒



$$\frac{32.5 \times 17.5}{32.5 + 17.5} = 11.25\ \Omega$$

⇒

Now,

$$I = \frac{V}{R}$$

$$= \frac{100}{36.25}$$

$$= 2.75 \, A$$